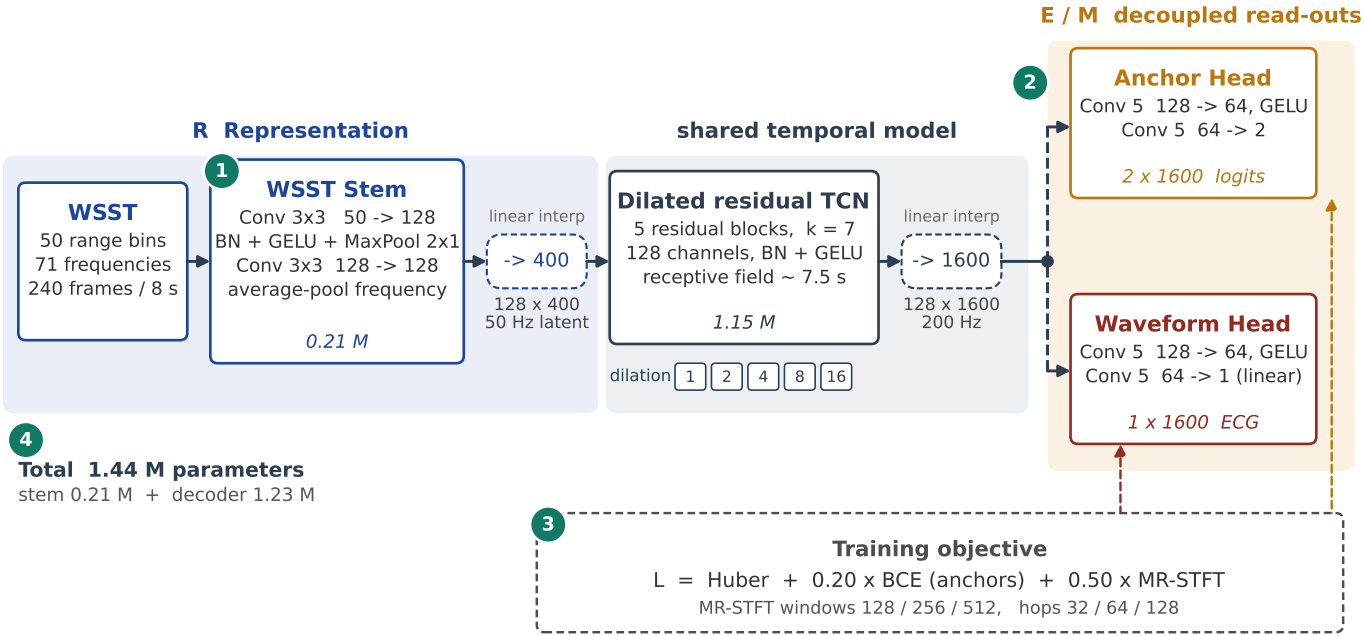

1 **Synchrosqueezed input.**
One stem over 50 range bins; frequency is pooled, time is kept.

2 **Factorized read-outs.**
Anchors and waveform are separate heads on one representation.

3 **Spectral objective (L3).**
Selected by the registered loss gate; median >10 Hz ratio 0.122.

4 **Lightweight.**
1.44 M against 41.6 M for the multitask ODE comparator.

Contributions are protocol- and design-level. No panel of this figure claims superior waveform morphology.